情報数学 I, 第 1 回レポート (4/21 締切) 解答例

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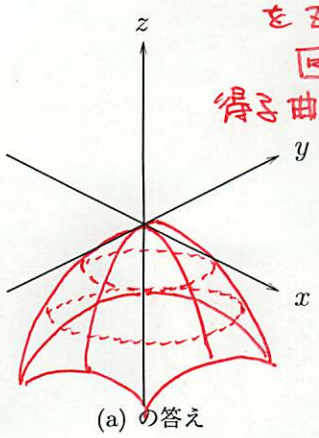
1 次の 2 変数関数 $z = f(x, y)$ のグラフの概形を描け.

(a) $z = -x^2 - y^2 \qquad (-1 \le x \le 1, -1 \le y \le 1)$

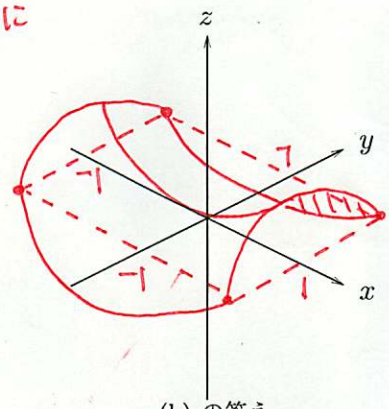
(b) $z = x^2 - y^2 \qquad (-1 \le x \le 1, -1 \le y \le 1)$

解)

放物線 ($z = -x^2$) を z 軸を中心に回転して得る曲面



(a) の答え



(b) の答え

馬の鞍

2 次の 2 変数関数 $z = f(x, y)$ の偏導関数 z_x, z_y および、2 階偏導関数 z_{xx}, z_{xy}, z_{yy} を求めよ.

(1) $z = x^3 + 2x^3y + y^4$

(2) $z = \cos(xy)$

解) (1) $z_x = \dfrac{\partial(x^3 + 2x^3y + y^4)}{\partial x} = 3x^2 + 6x^2y, \qquad z_y = \dfrac{\partial(x^3 + 2x^3y + y^4)}{\partial y} = 2x^3 + 4y^3,$

$$z_{xx} = \frac{\partial(3x^2 + 6x^2y)}{\partial x} = 6x + 12xy, \qquad z_{xy} = \frac{\partial(3x^2 + 6x^2y)}{\partial y} = 6x^2, \qquad z_{yy} = \frac{\partial(2x^3 + 4y^3)}{\partial y} = 12y^2.$$

(2) $z_x = -y\sin(xy), \qquad z_y = -x\sin(xy),$

$$z_{xx} = \frac{\partial(-y\sin(xy))}{\partial x} = -y^2\cos(xy), \qquad z_{yy} = \frac{\partial(-x\sin(xy))}{\partial y} = -x^2\cos(xy),$$

$$z_{xy} = \frac{\partial(-y\sin(xy))}{\partial y} = \frac{\partial(-y)}{\partial y}\sin(xy) + (-y)\frac{\partial(\sin(xy))}{\partial y} = -\sin(xy) - xy\cos(xy).$$

[0]※この講義に関する情報は FTP フォルダを参照. ftp://ftpi.sie.dendai.ac.jp/pub/nasu/Information Mathematics I/